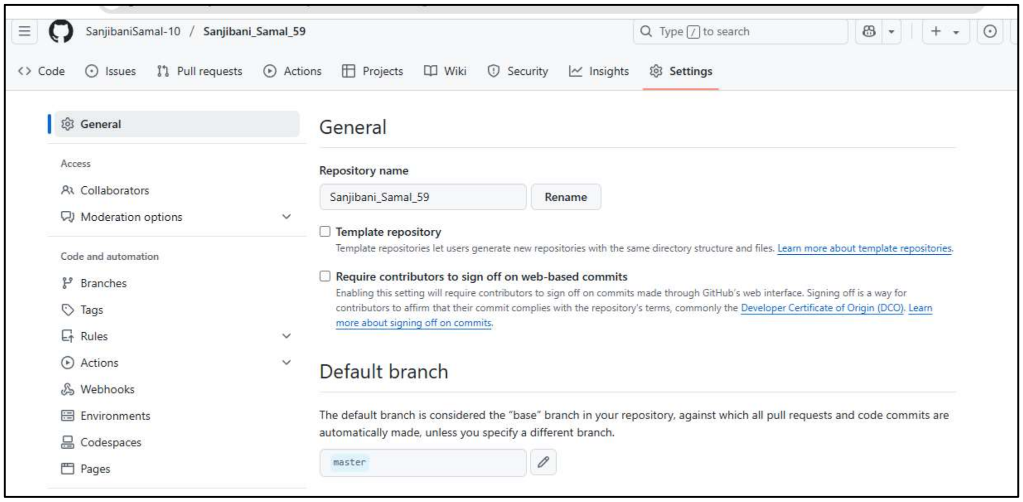
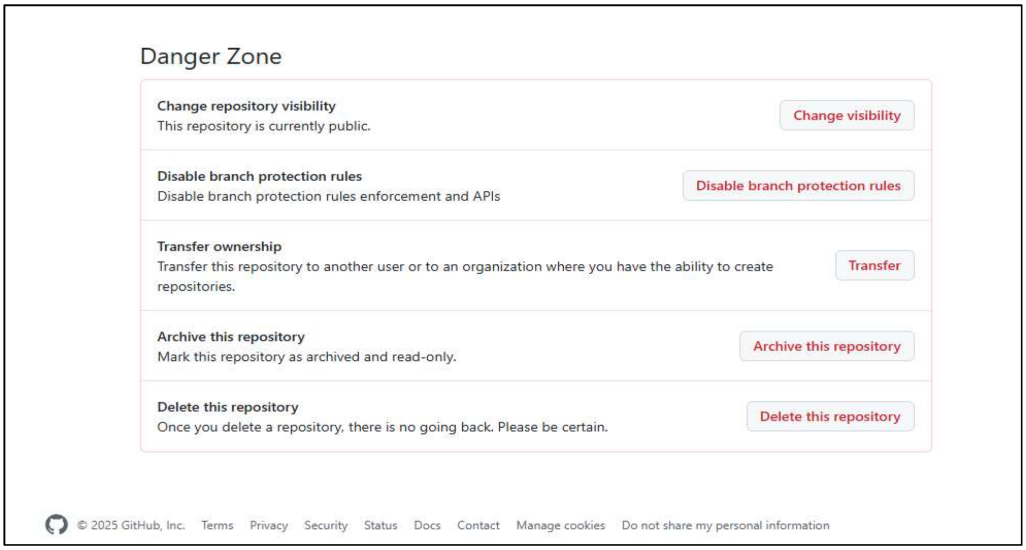
ASSIGNMENT NO :- 11

TITLE :- Build scaling plans in AWS that balance the load on different EC2 instances.

STEP 1 :- Check if your GitHub Repository is public or not . For that Login to your **GitHub Account** and then select your Repository and then go to **Setting** and navigate to **Danger Zone** and check your **Repository Visibility.**





STEP 2 :- Login to your **AWS Account** and click on **EC2** and go to **Security Groups** and click on **Create Security Group**

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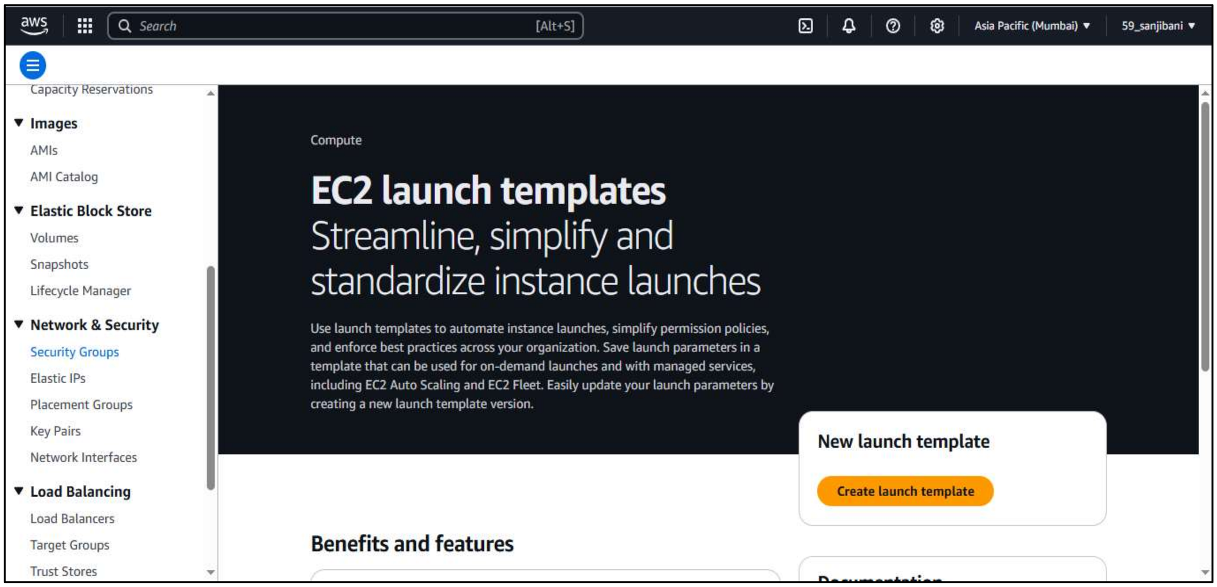
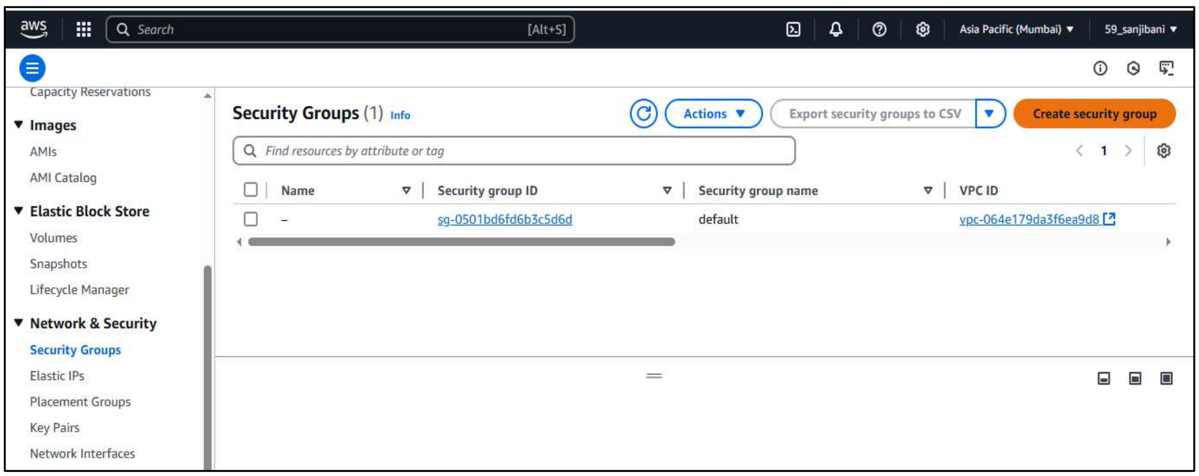
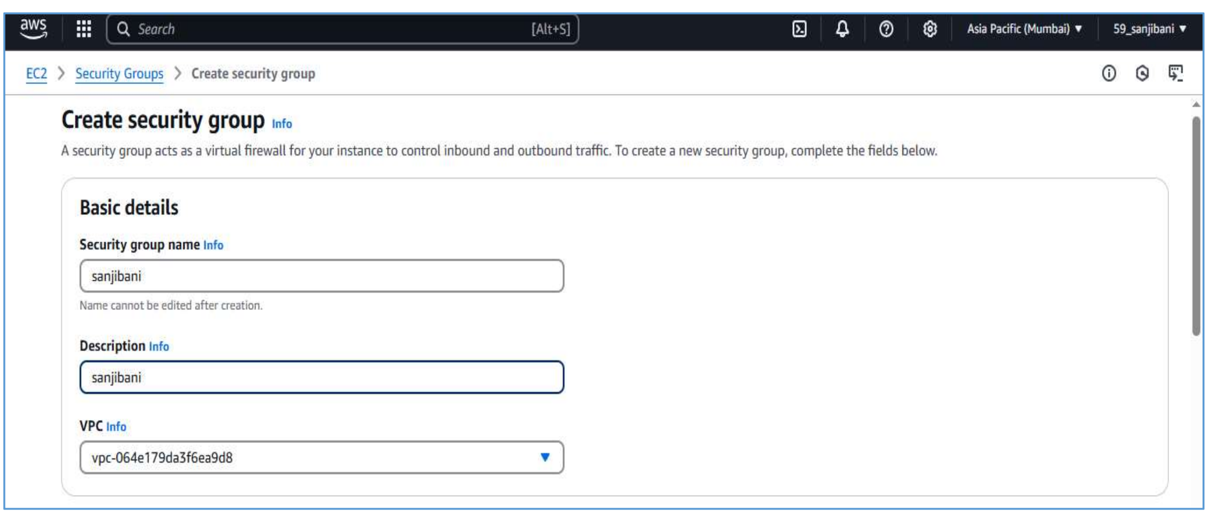




>> Give the **Security Group Name** and **Description** and the click on **Add Rules** of Inbound Rules



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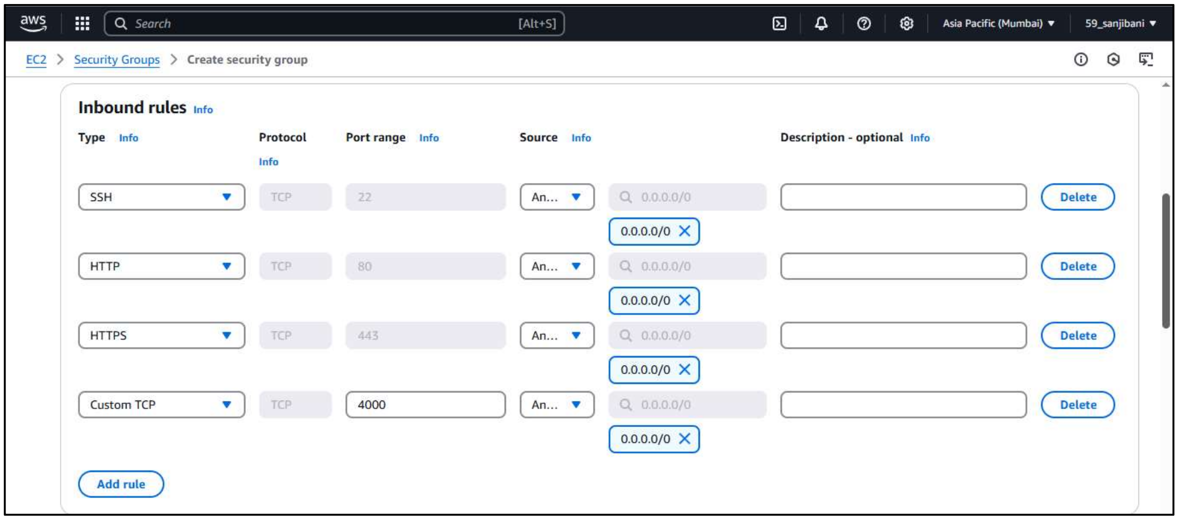
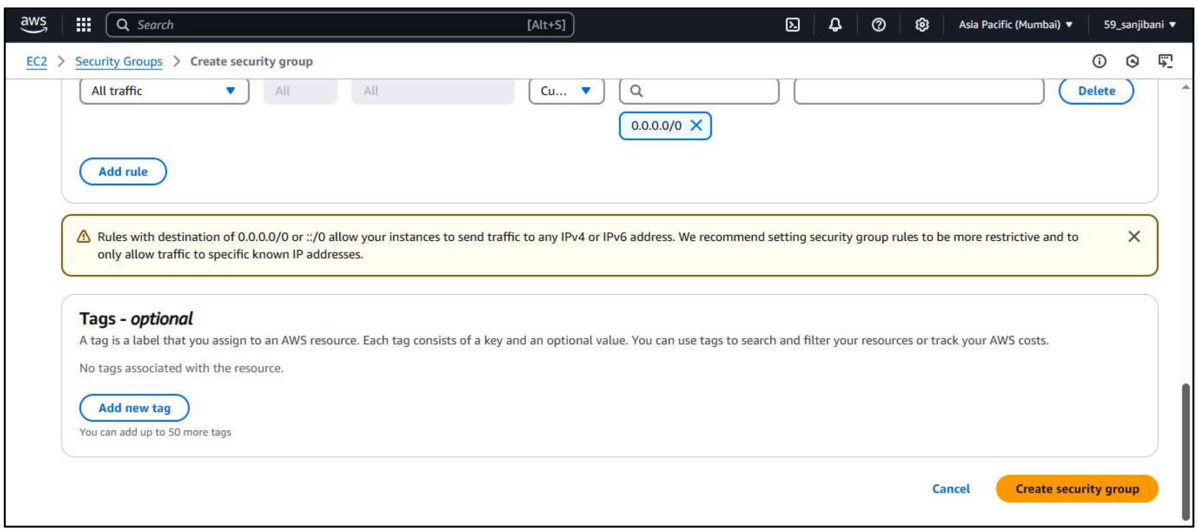
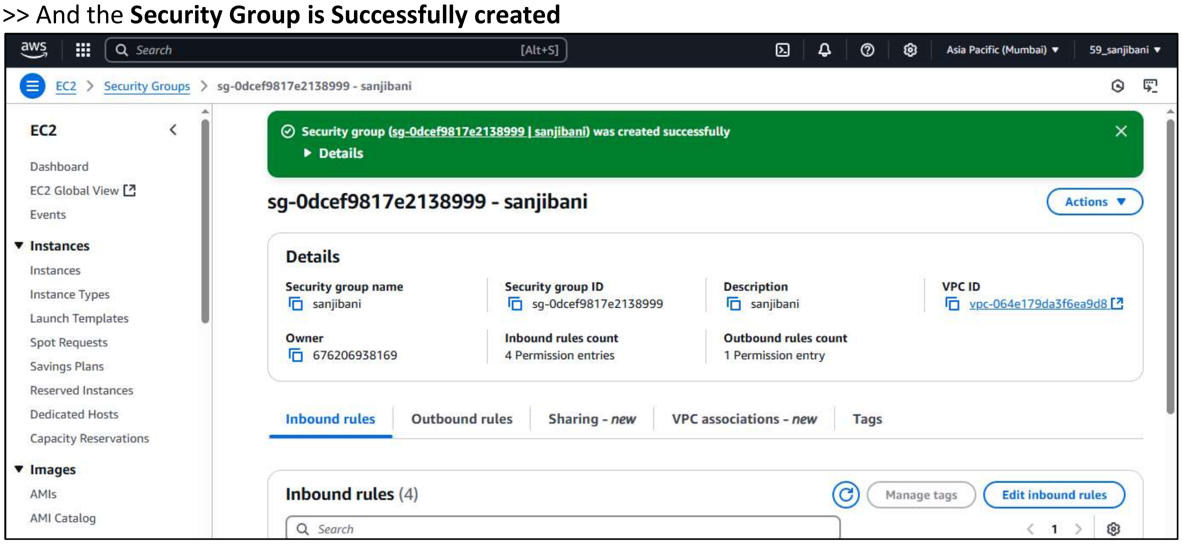
>> Add **SSH** , **HTTP** , **HTTPS** and **CUSTOM TCP** Type Rules as Given below



>> Click on **CREATE Security Group**



>> And the **Security Group is Successfully created**



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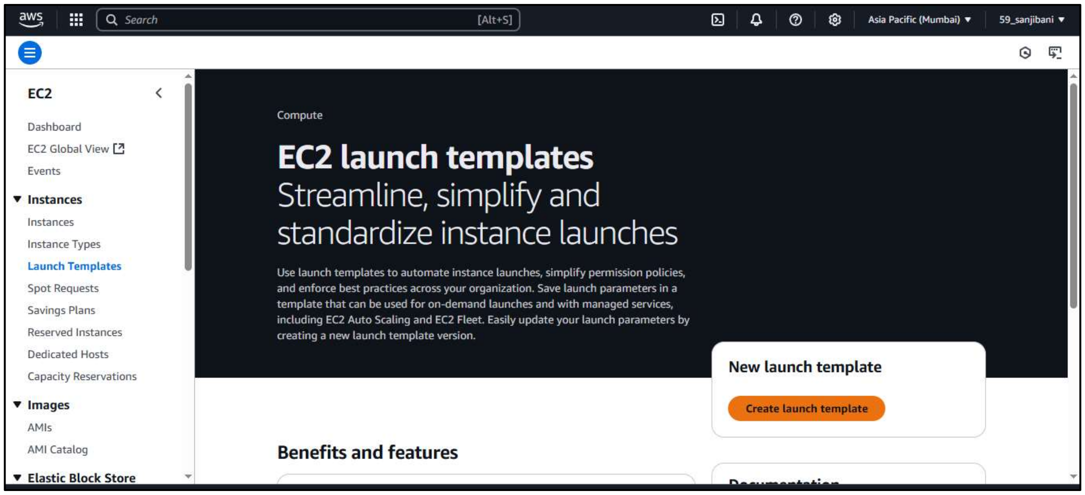
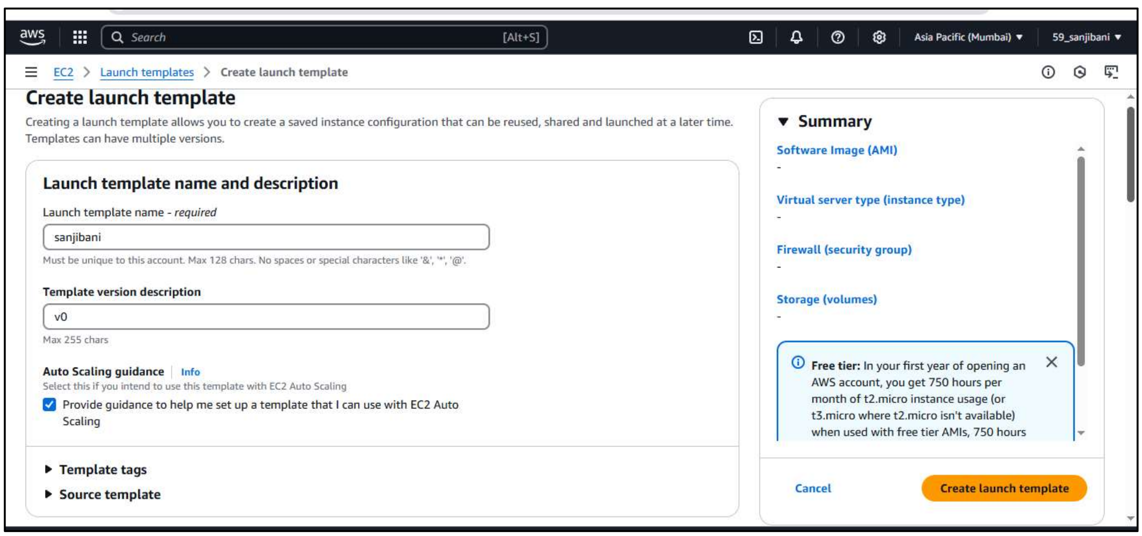
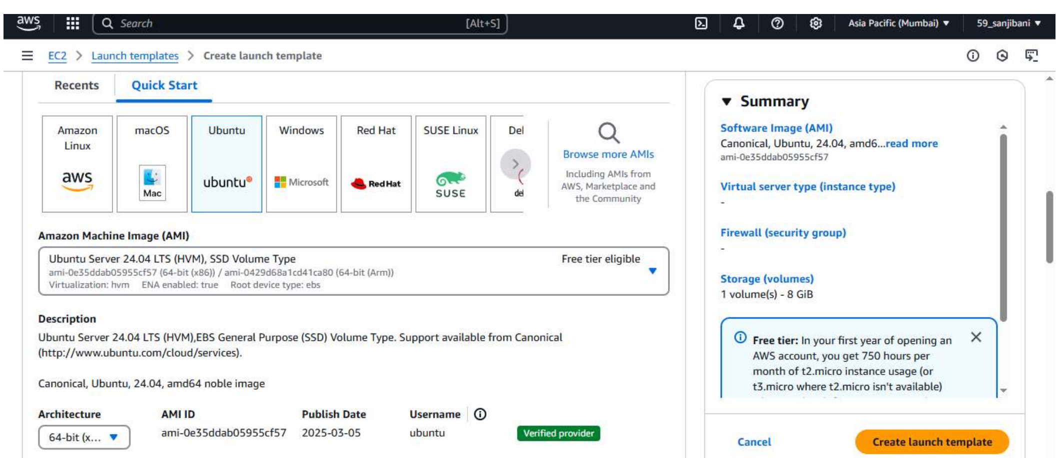
STEP 3 :- Then go to **Launch Template** and click on **Create Launch Template**



STEP 4:- Give your **Template name** and **version description** and select **Auto Scaling guidance**



>> Select **Ubuntu** in Quickstart



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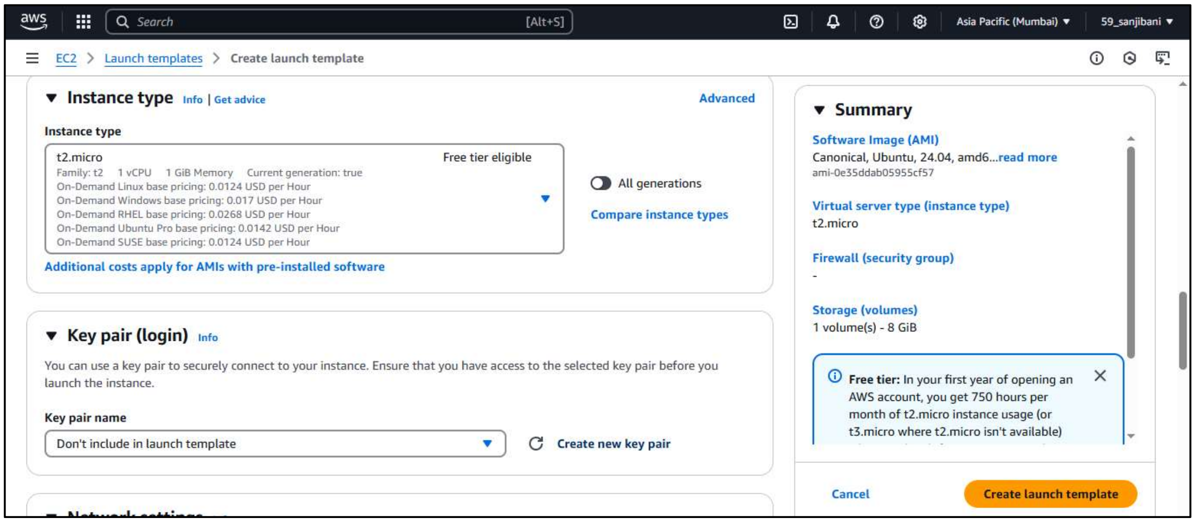
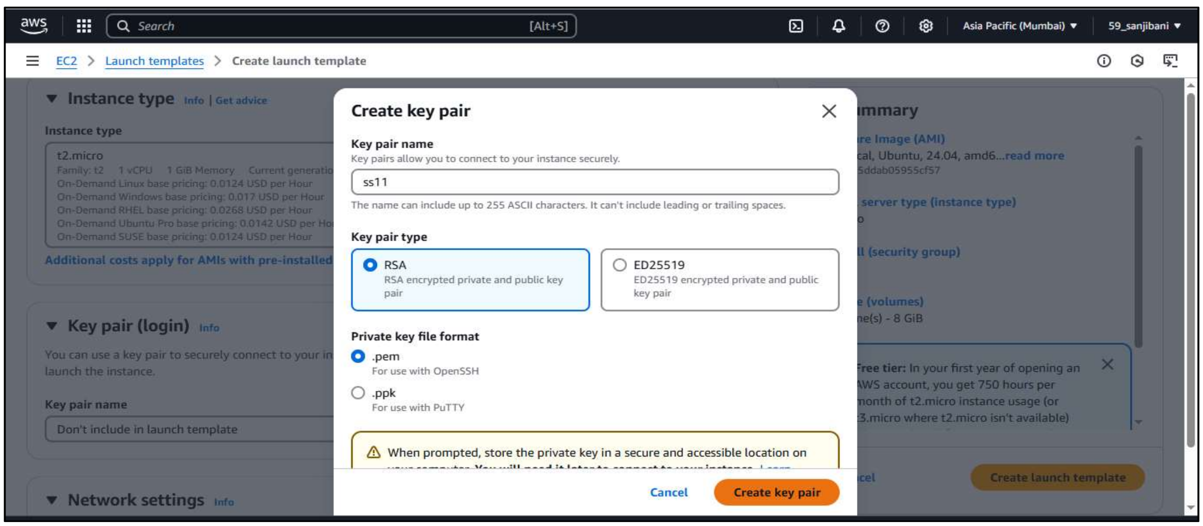
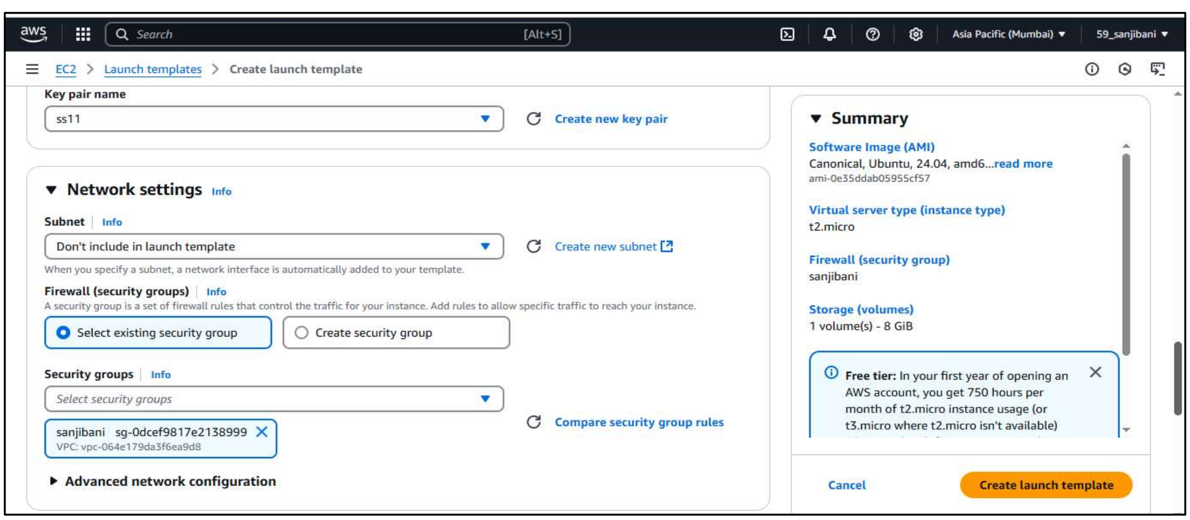
>> Select **t2.micro** as **Instance type** and create a **key pair**





>> Select the **Security group** you created



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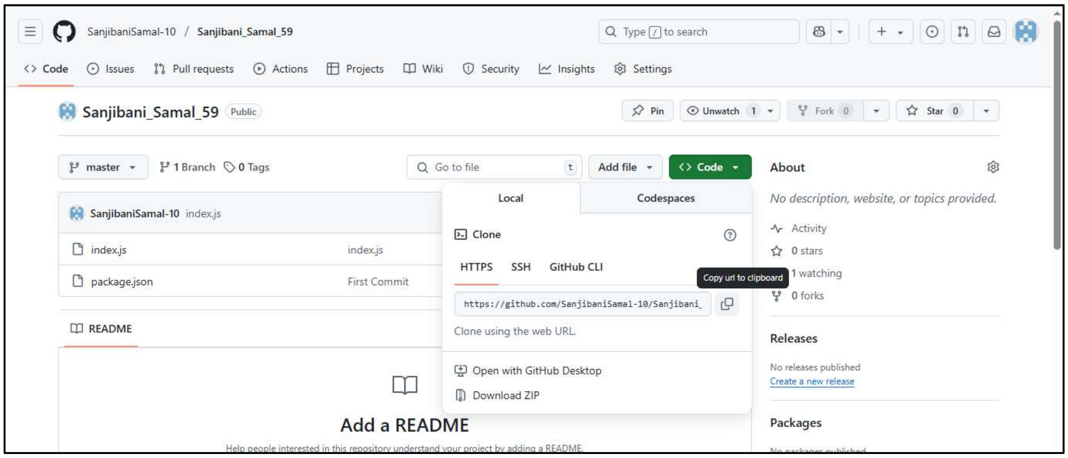
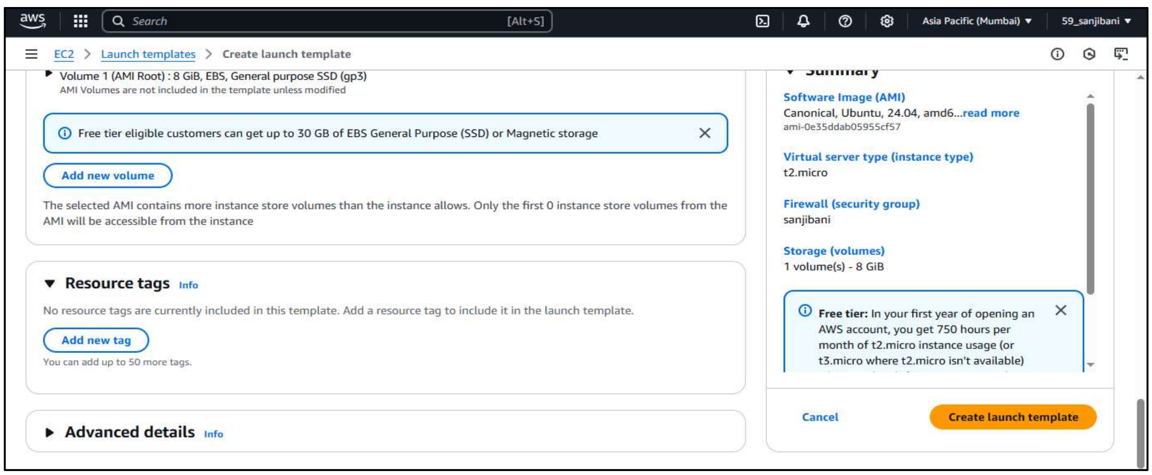
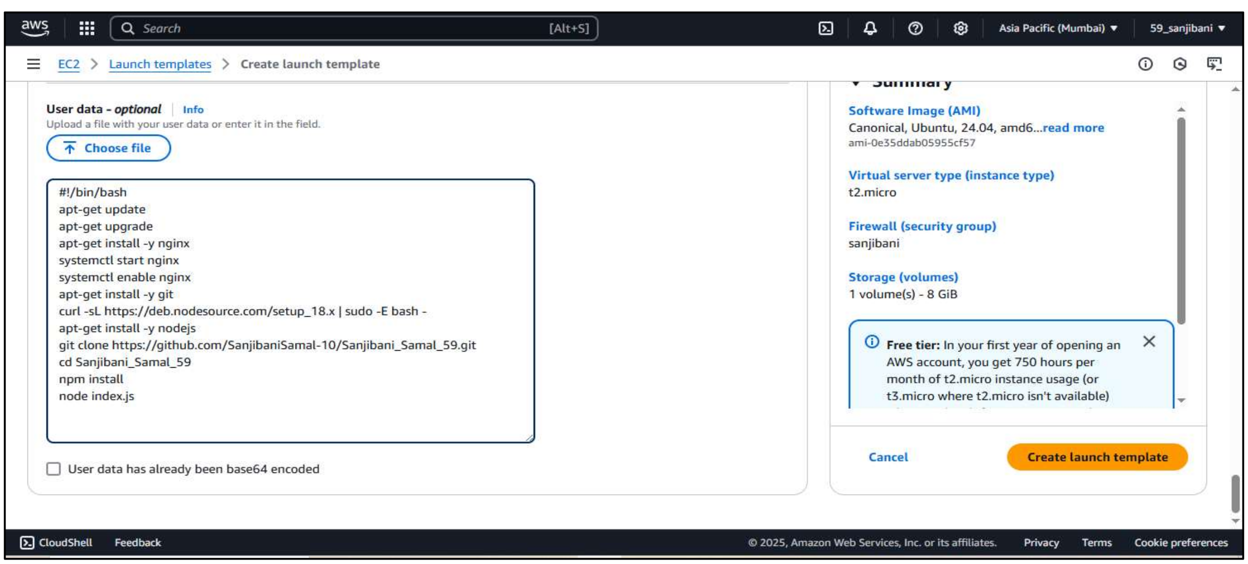
>> Then go to your **Github Repository** and copy the address .



>> then again move to the template createion and click on **Advanced Details**



>> write the command line as written below and add your Repository link in **Git Clone <Repository Link>**



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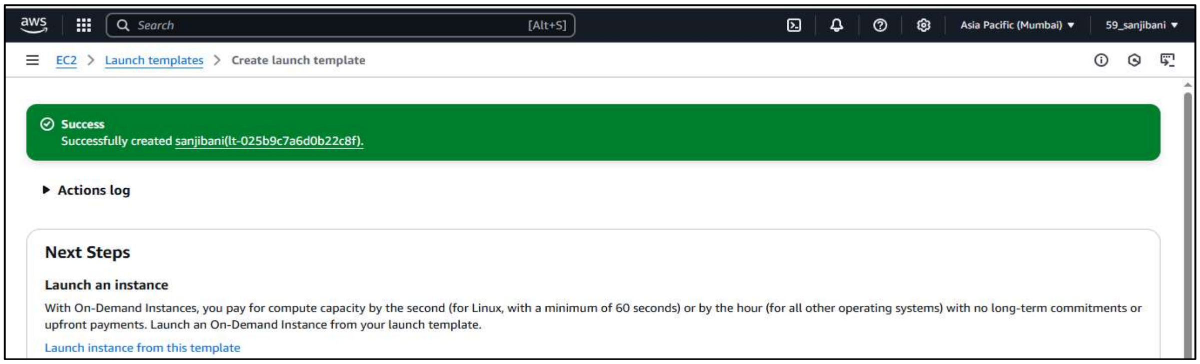
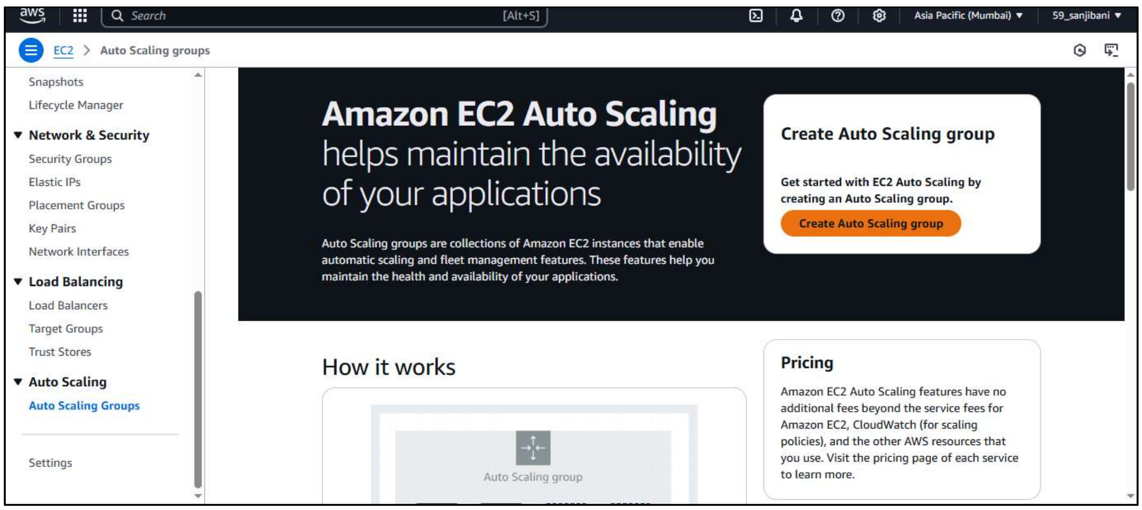
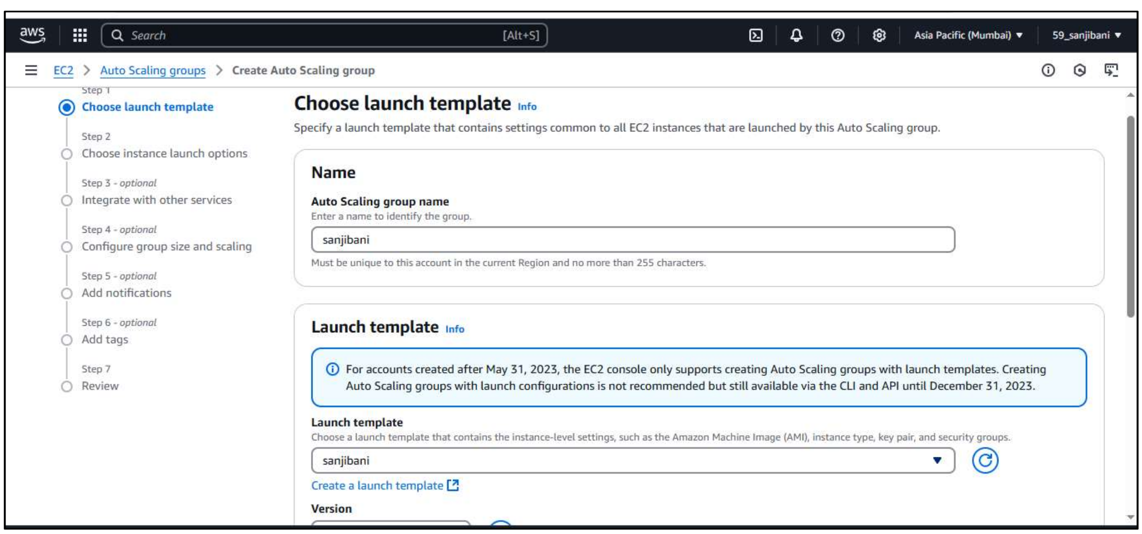
>> Your Template is created successfully.



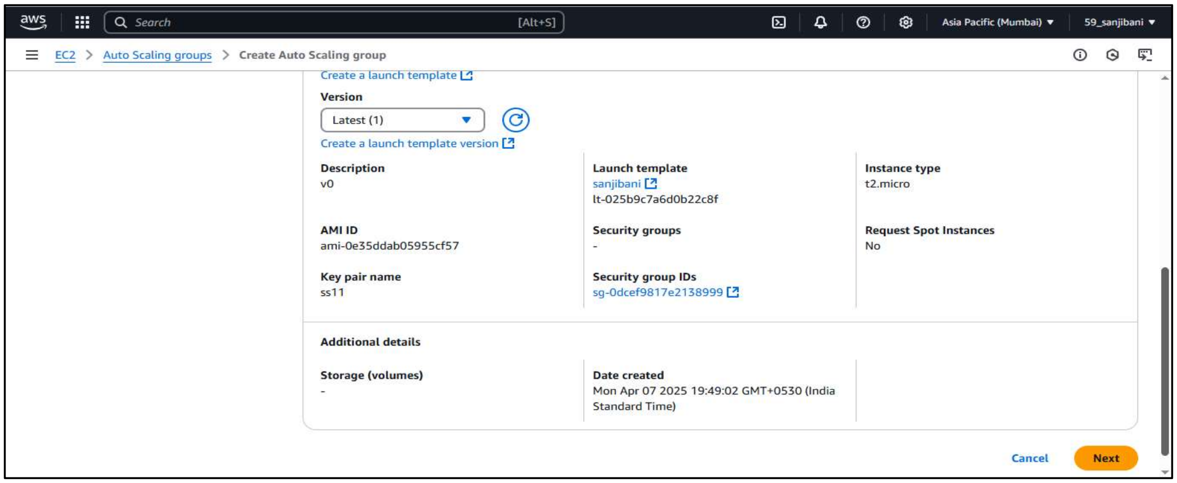
STEP 5:- Click on **Auto Scaling Groups** and then click on **Create Auto Scaling groups.**



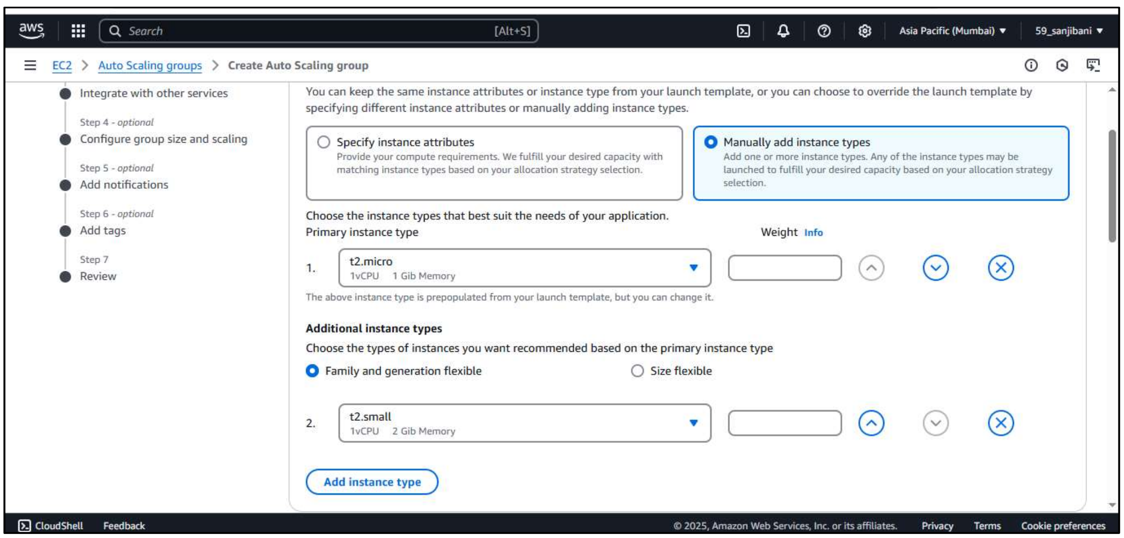
>> Give a **name** and select your desired **Template** and click on Next



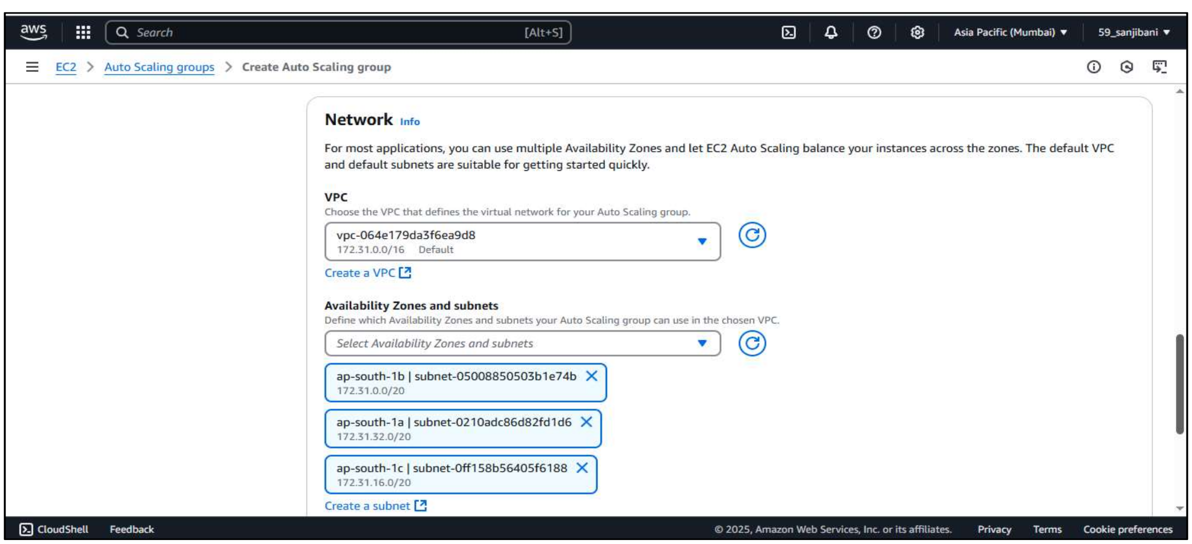
>> Give the version as **Latest (1)** and click on Next



>> Select the **Manual add Instance type**



>>In Network region Select the Three Available Zones as servers and click on next



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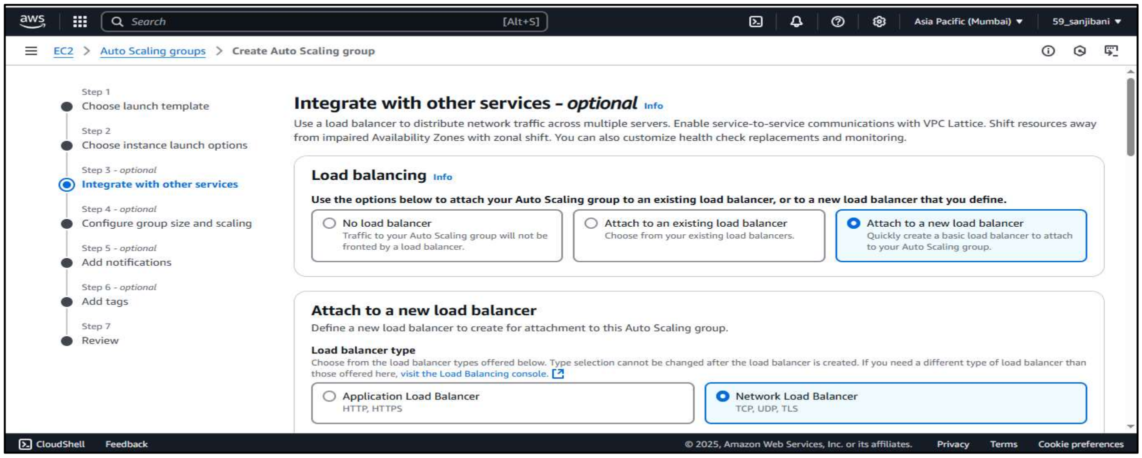
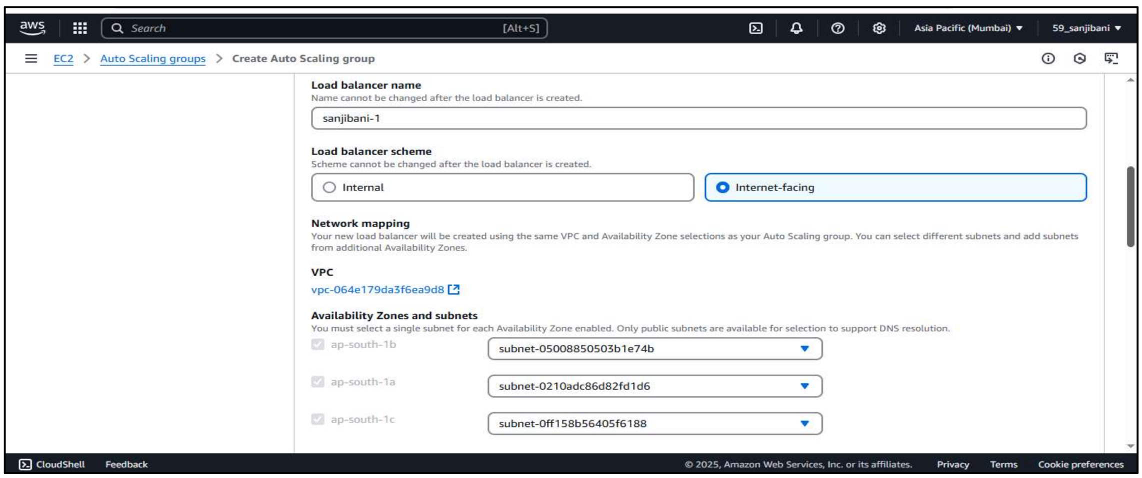
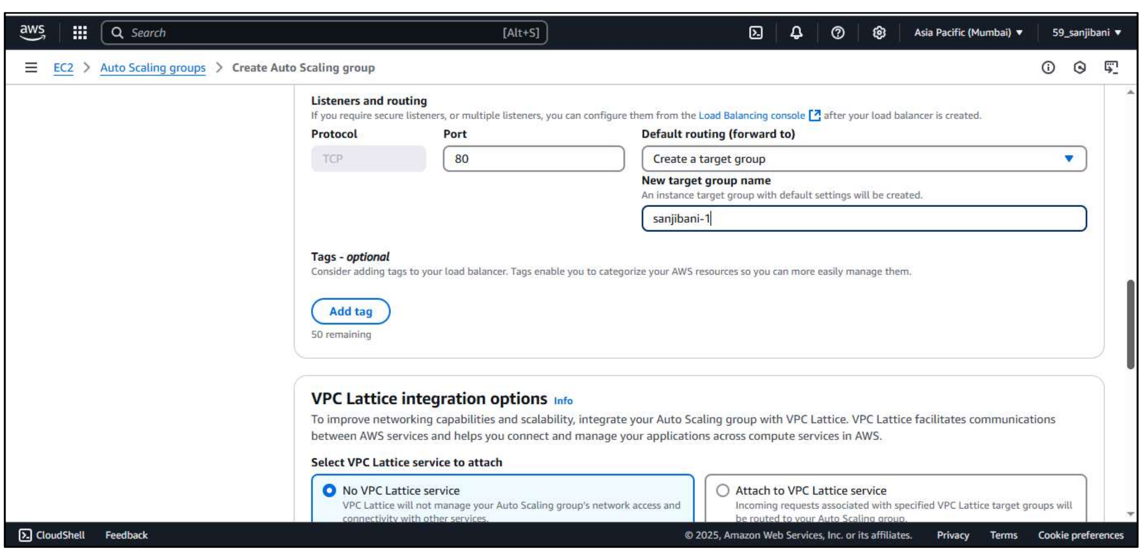
>> Select **Attach a new load balancer** and then select **Network load balancer** and then select **Internet Facing**





>> click on Create Target Group and select a **target group name** and then click on next



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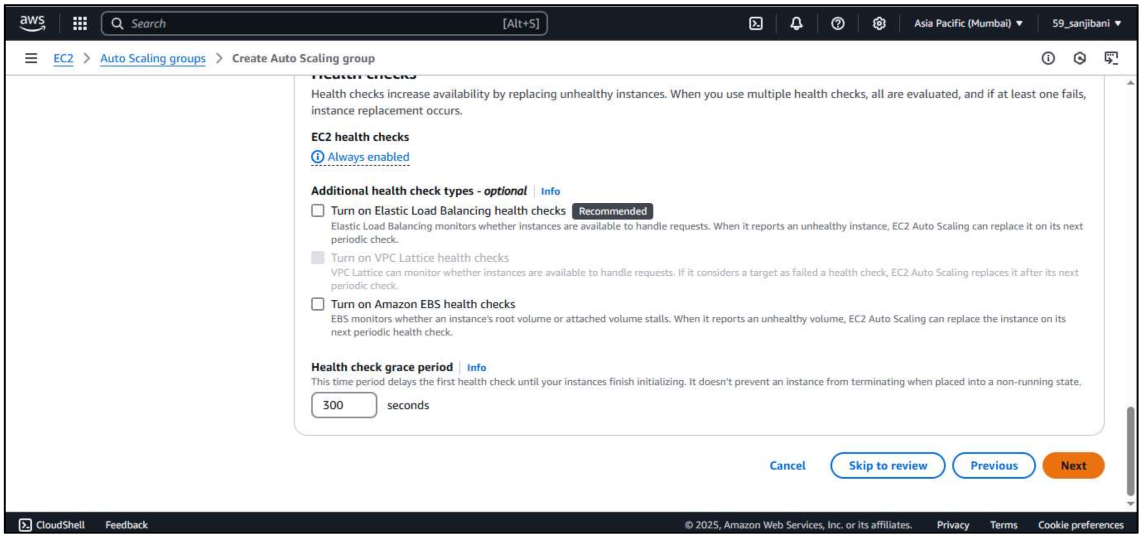
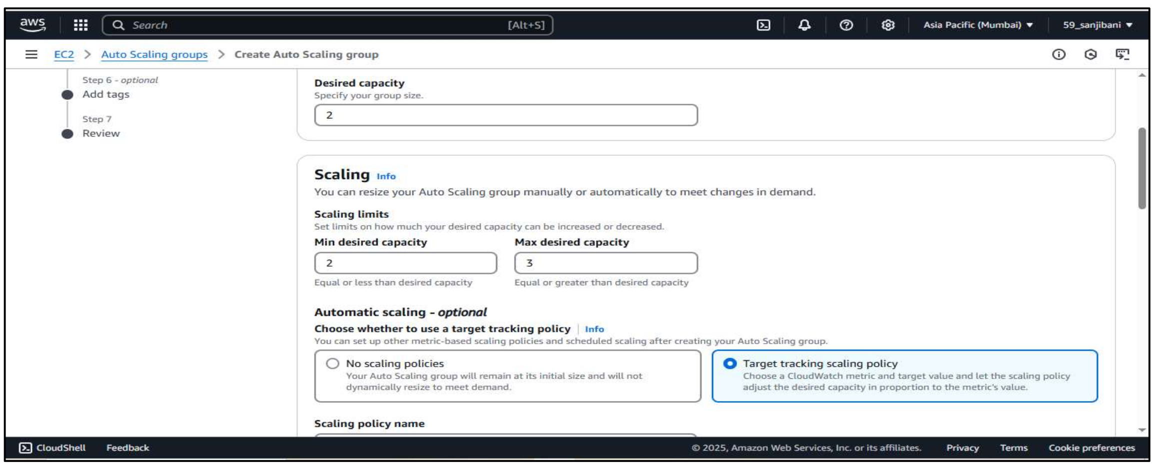
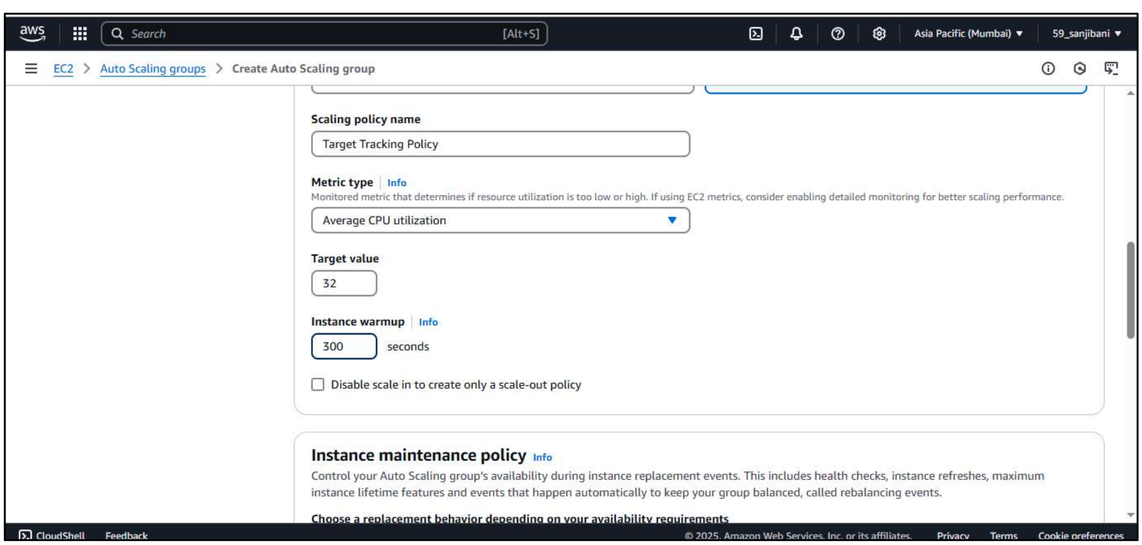


>> Now select your **Desired capacity** , **Min desired and max desired capacity** and then select **Target Tracking Scaling capacity.**



>> Give the **Target Value** and give **Instance Runup** and then click on **Next**



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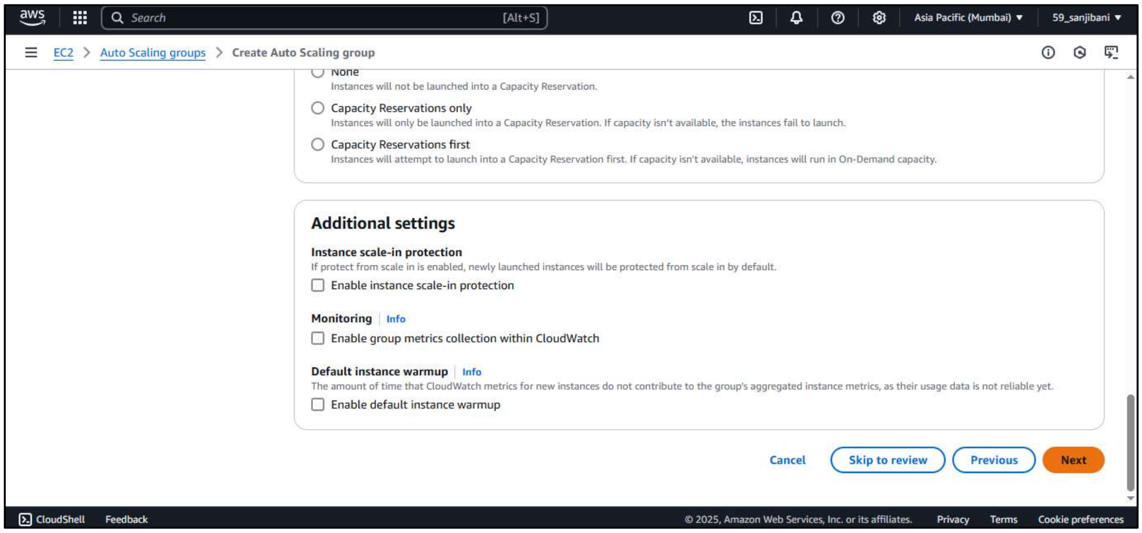
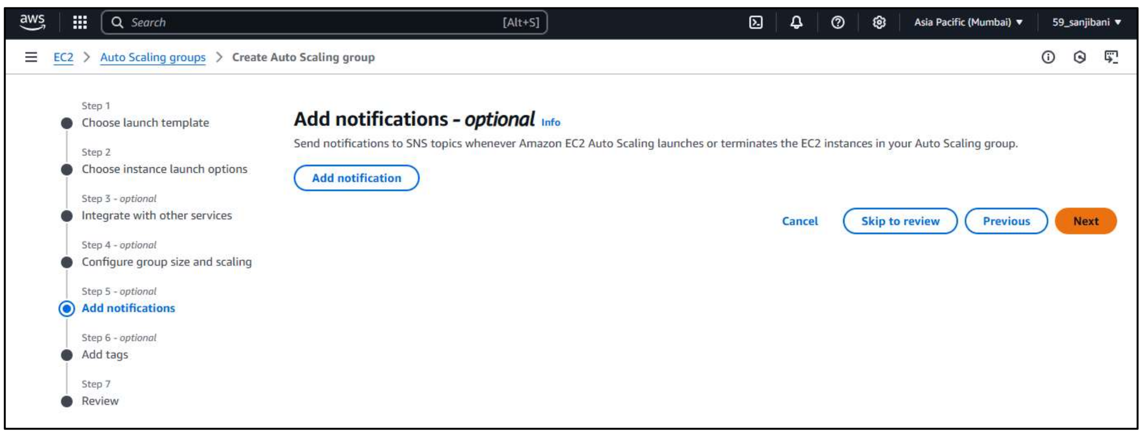
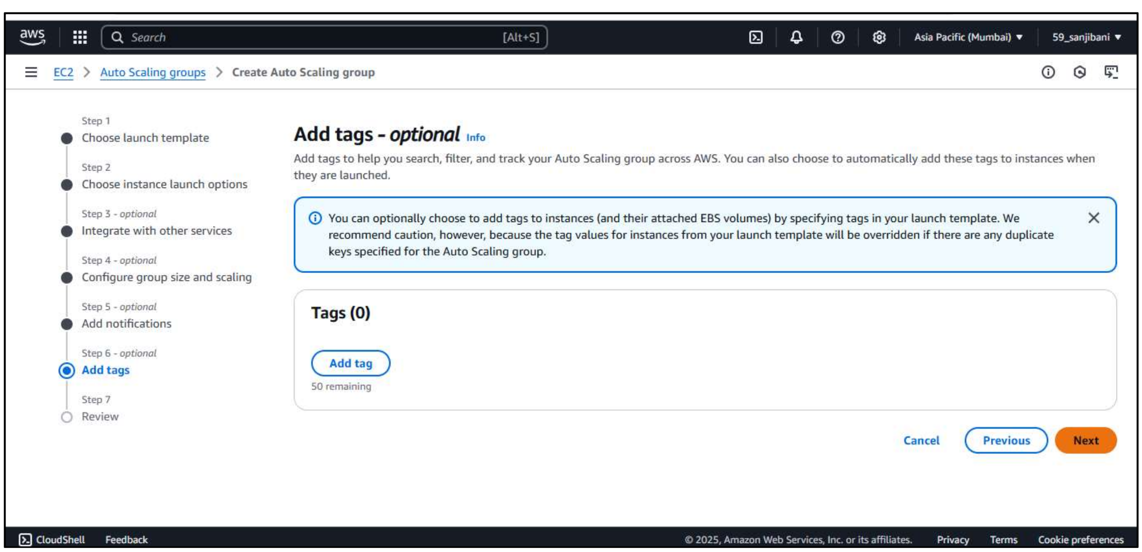


>>click on **NEXT**



>>click on **NEXT**



>>click on **Create Auto Scaling Group**

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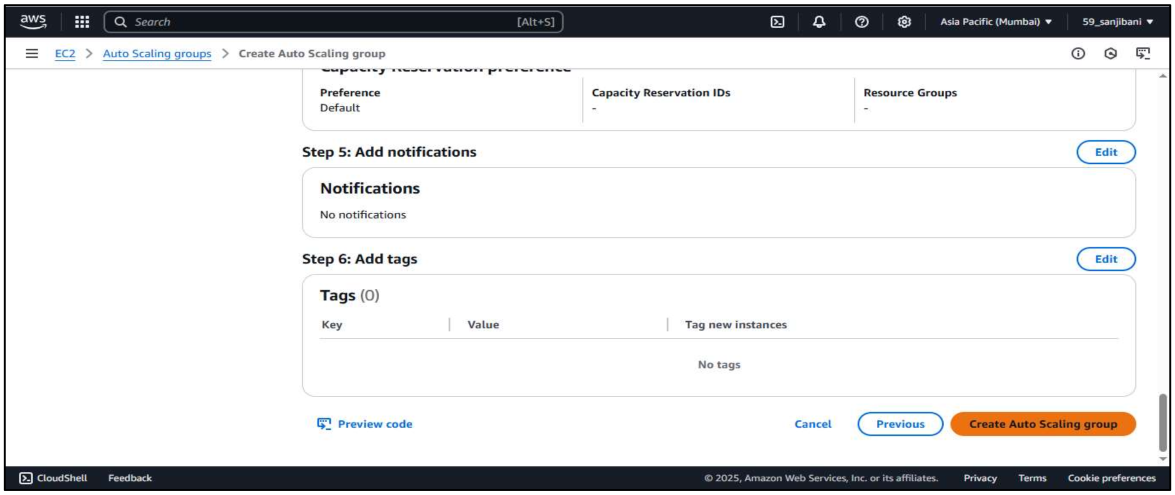
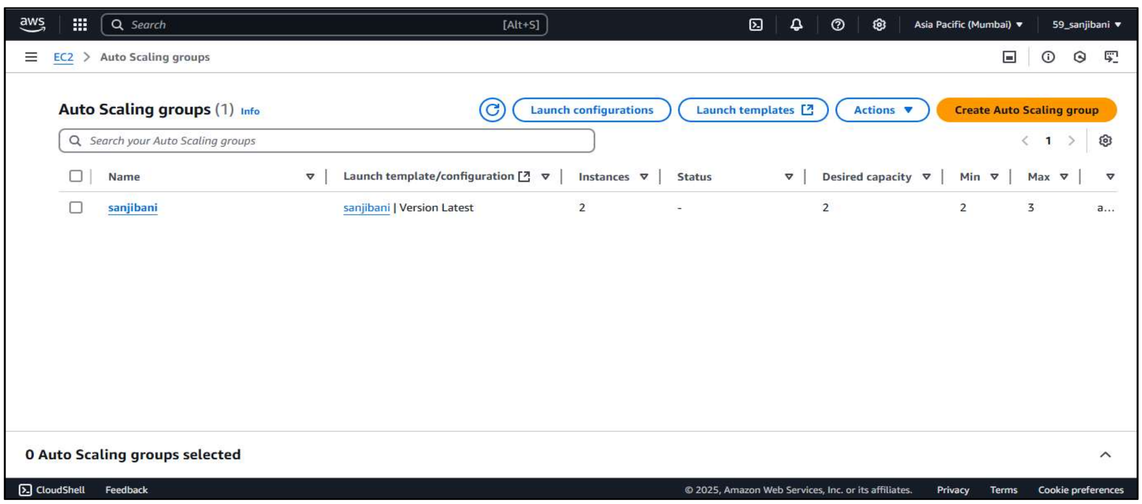
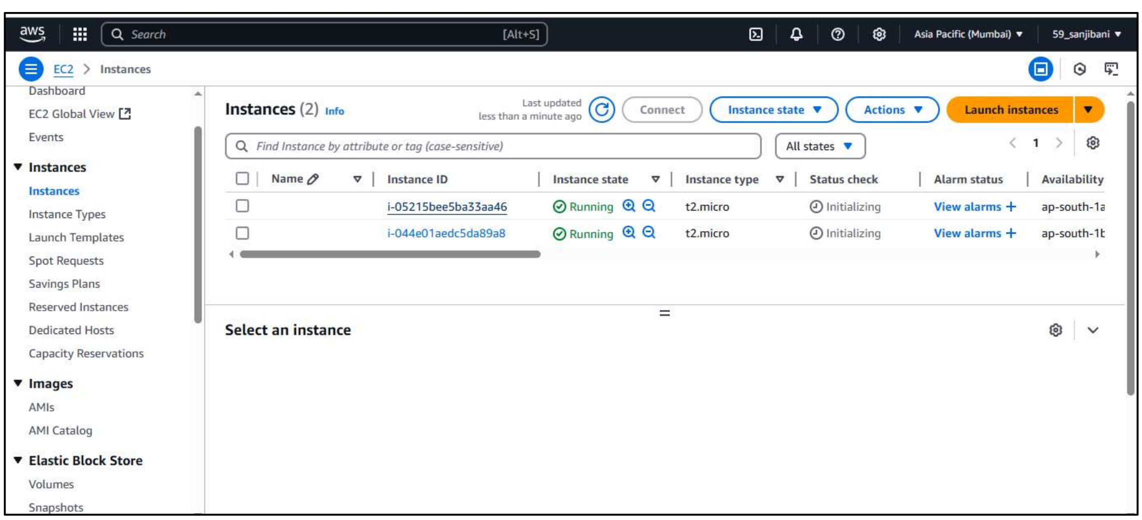


>> And your Auto Scaling Group is Created Successfully



STEP 5:- As you can see the **Instances are automatically created** if you will delete one instance then the 3rd one will automatically generated as **3** servers are given



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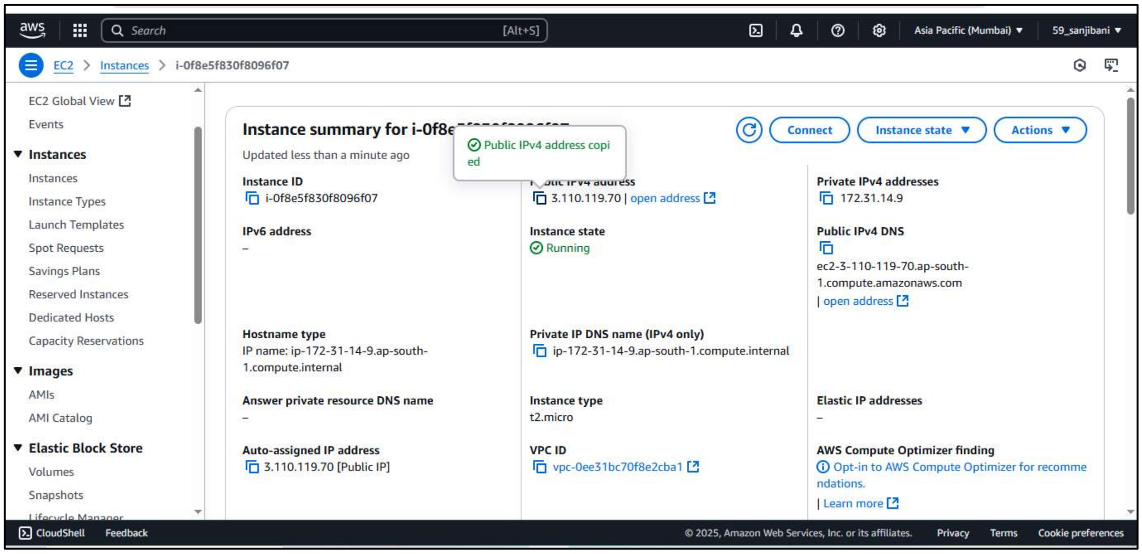
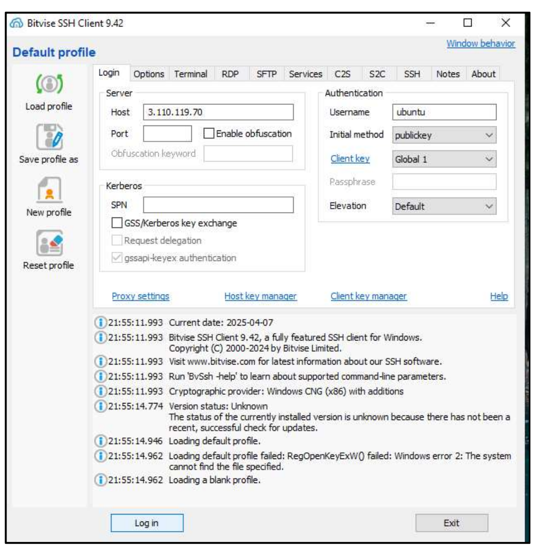
STEP 6:- Select any one **Instance** and Copy its **IPV4 address** and paste it in **Incognito website** with port number(**IP Address : 4000**)



STEP 7:- Paste the address in **Bitvise Server** and give the required field of **username , Initial method** and **Client key** then click on **LOG IN** and go to the **New Terminal**



>> Then write the following command in terminal to open a shell file

```
ubuntu@ip-172-31-39-192:~$ nano infil.sh
```

>> Then write the following code in the shell file

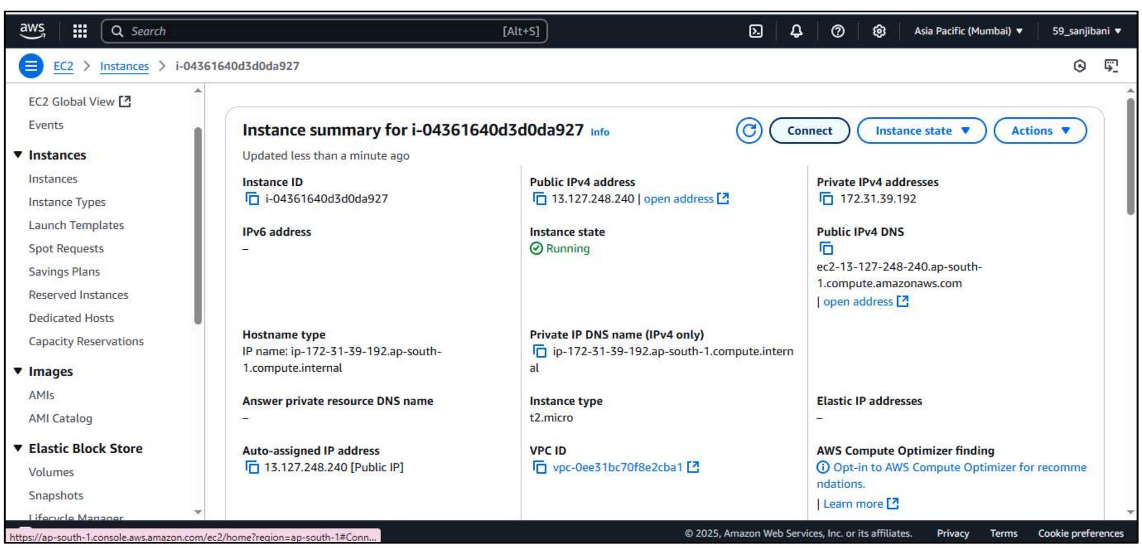
```
GNU nano 7.2                                    infil.sh *
#!/bin/bash
while true
do
echo "looping"
done
```

>> Compile and run the File Using the following command

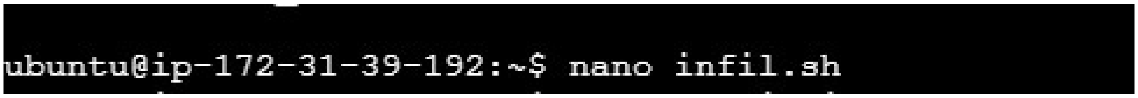
```
ubuntu@ip-172-31-39-192:~$ nano infil.sh
ubuntu@ip-172-31-39-192:~$ chmod +x infil.sh
ubuntu@ip-172-31-39-192:~$ ./infil.sh
```

```
ubuntu@3.110.119.70:22 - Bitvise xterm - ubuntu@ip-172-31-14-9: ~
looping
looping
looping
looping
looping
looping
looping
looping
looping
looping
looping
looping
looping
looping
looping
looping
looping
looping
looping
looping
looping
```

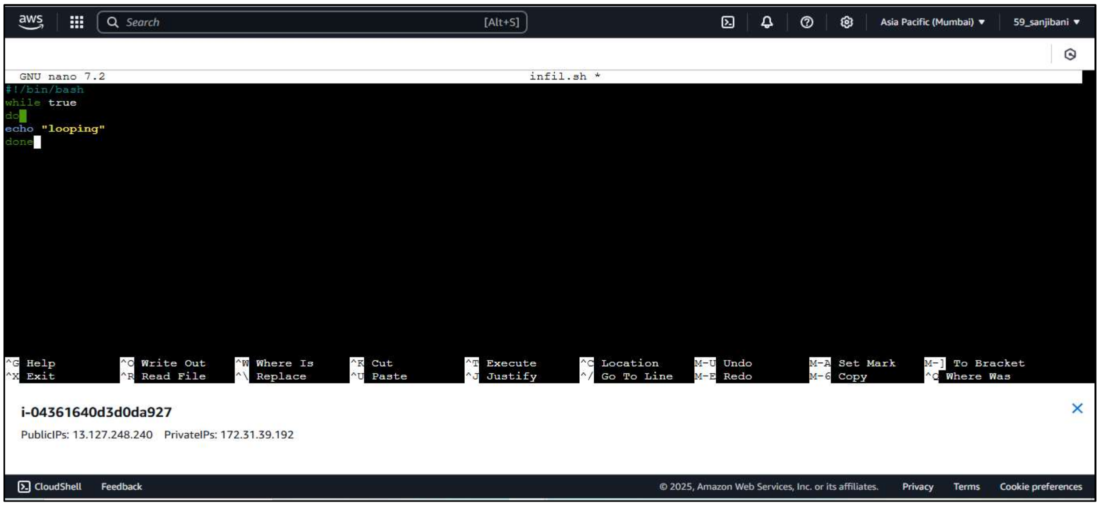
<u>STEP 6:-</u> Select other **Instance** and click on **Connect**



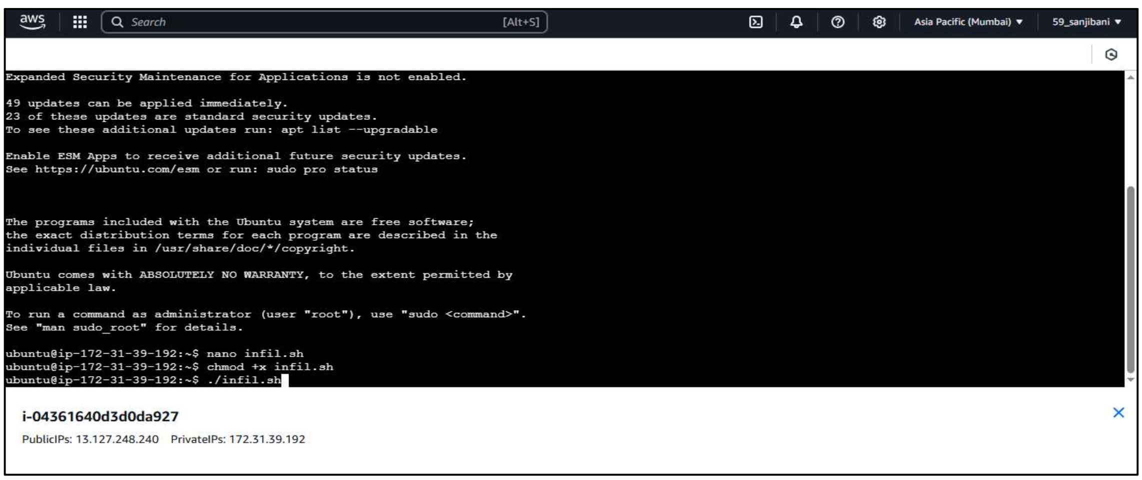
>> Then write the following command in terminal to open a shell file

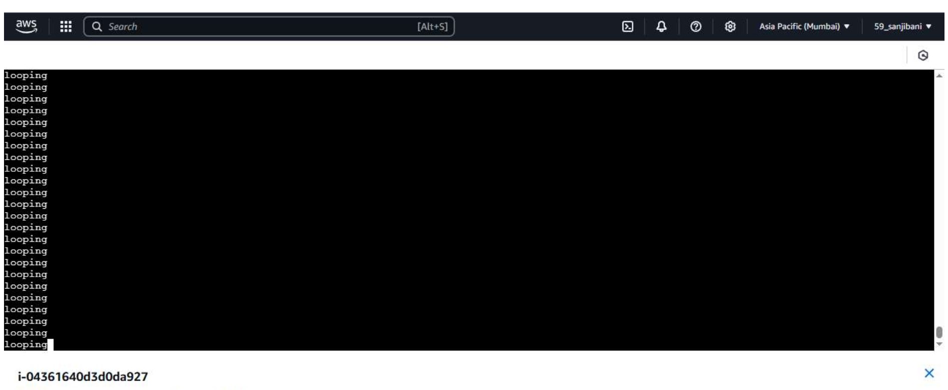


>> Then write the following code in the shell file



>> Compile and run the File Using the following command





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>> Then click on both the instances and you can see the graph of both processes **CPU UTILISATION**